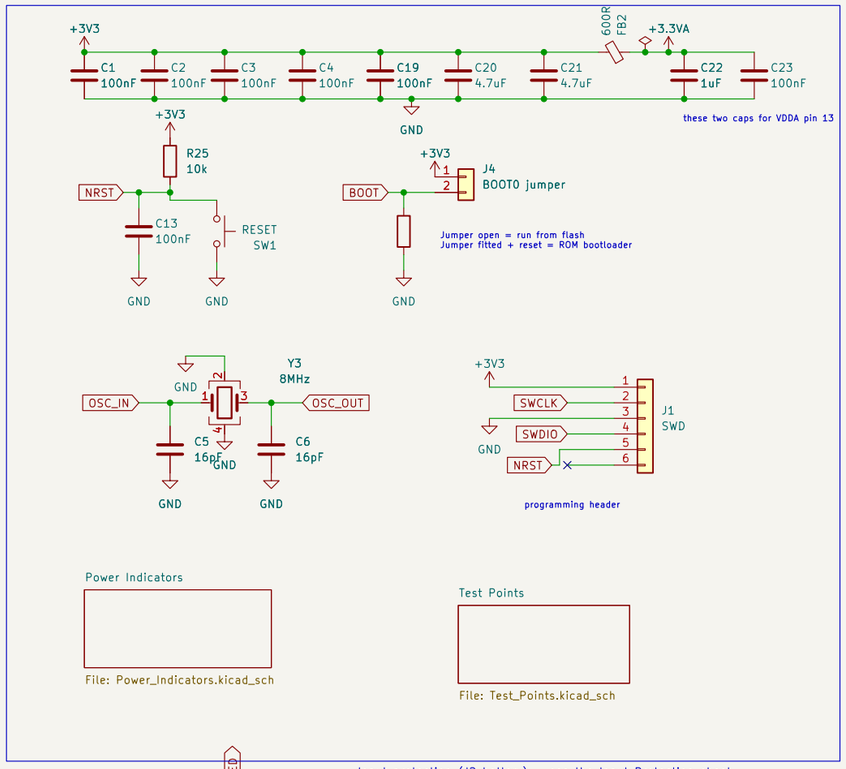
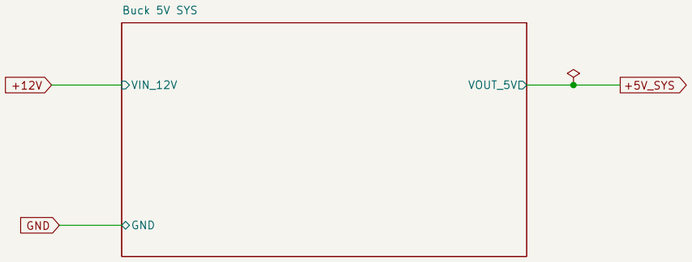




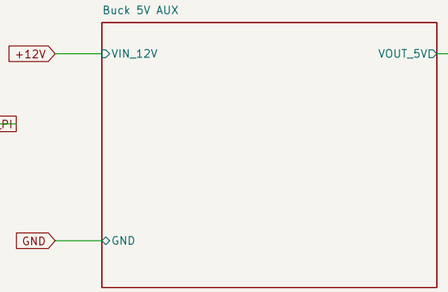


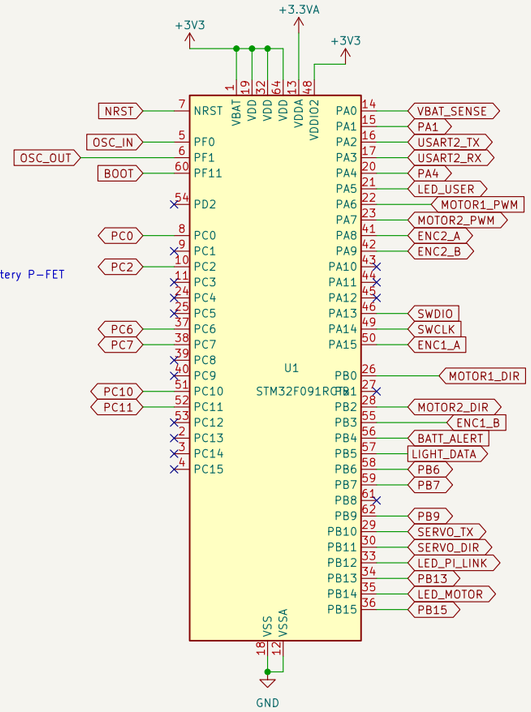
Input protection (J2 battery) — see the Input Protection sheet
J2 XT30 -> VBAT_IN -> F1 10 A MINI fuse -> VBAT_FUSED -> Q1 reverse-battery P-FET -> +12V (D13 SMBJ16A TVS, C36 bulk, battery sense on PA0).
C36 is on +12V after Q1, so it is never reverse-biased.
Battery harness fuse at the M8 lugs: 15 A.

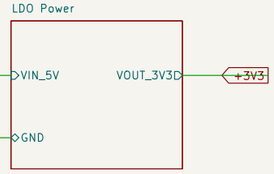


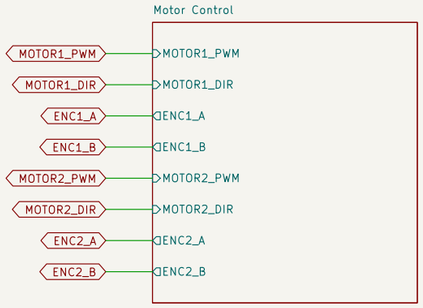


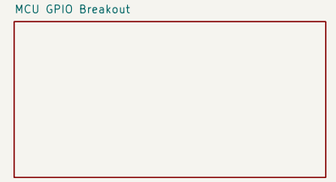


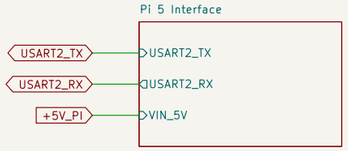


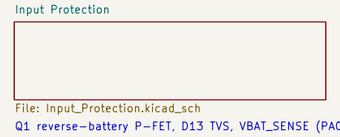



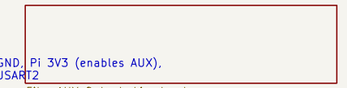













Mounting holes — 4x M3, 3.2 mm non-plated (NPTH), no copper, no net.
Place one per board corner; not in the BOM (no part to buy).




ECE49022 Team 22 senior design — STM32F091 control, power and I/O
Derived from ST NUCLEO-64 MB1136 rev C-04, sheets 2–3

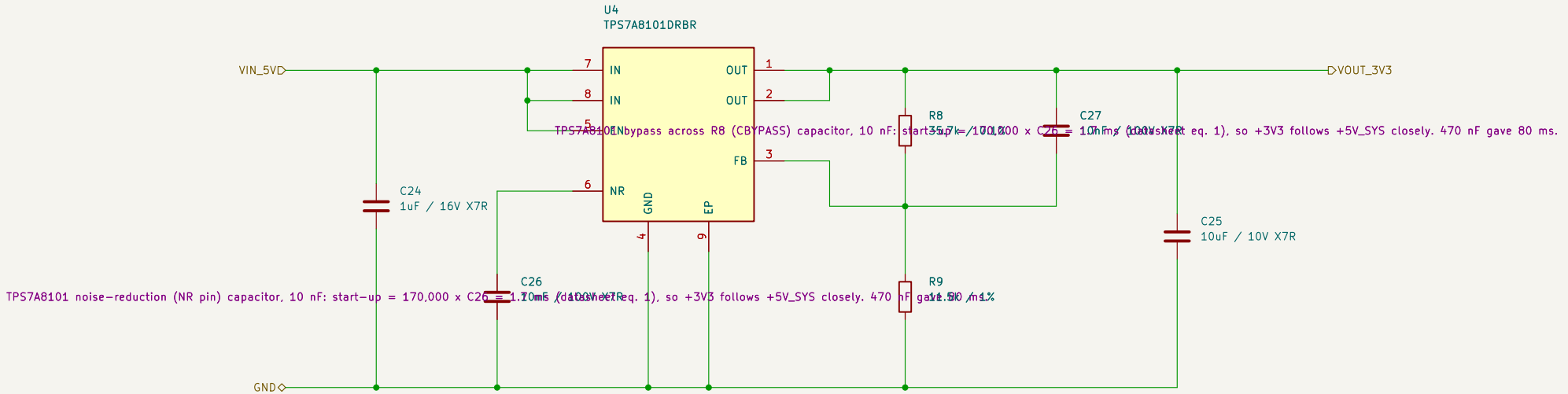
Sheet: /
File: delivery_robot_1.kicad_sch

**Title: Delivery Robot Main Board**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-08 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 1/15 |

# 5 V to 3.3 V LDO supply

Reusable sheet: VIN_5V input, VOUT_3V3 output, common GND




VOUT nominal = 0.8 x (1 + 35.7k / 11.5k) = 3.284 V (E96 values)
1 A is the TOTAL output budget for the STM32 and other 3.3 V loads.

C25: require >= 4.7 uF EFFECTIVE capacitance; selected part modeled at 8.8 uF.
EN tied to IN. C26 (NR) and C27 (bypass across R8) are 10 nF: start-up about 1.7 ms
(TI tests the part with CNR = 10 nF, CBYPASS = 0). 470 nF gave 80 ms, during which
5 V encoder and servo signals could reach MCU pins while VDD was still 0.

Layout: capacitors close to U4; EP pad 9 to GND copper and thermal vias.
Heat: the real 3.3 V load is about 150 mA, about 0.25 W at 5 V input
(1.7 W only at the full 1 A rating).

5 V motor / encoder supply branches BEFORE this sheet.

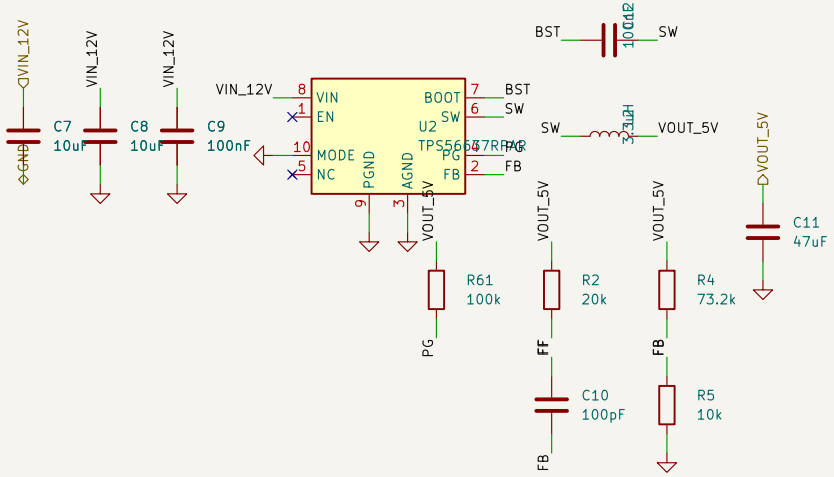
5 V to 3.3 V, 1 A total rail capacity

**Senior Design**

Sheet: /LDO Power/
File: LDO_3V3.kicad_sch

***Title: LDO power supply — reusable sheet***

| Size: A4 | Date: 2026-10-05 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/15 |



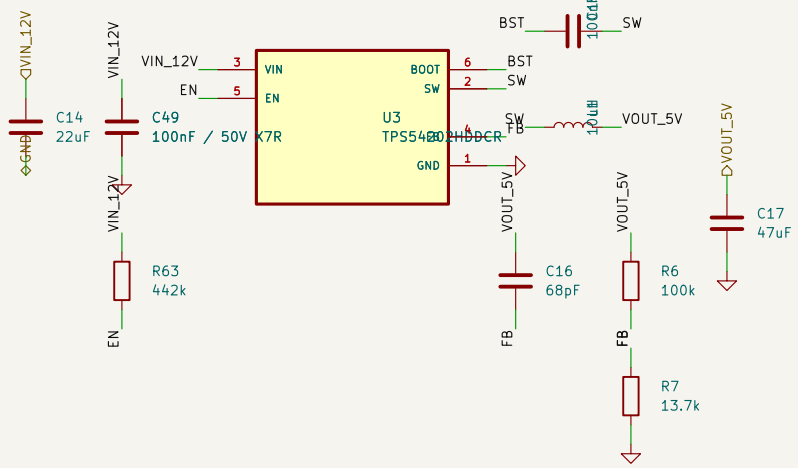

TPS56637 (U2) 12 V to 5 V 6 A, values from TI WEBENCH (EN floating = enabled)

**Senior Design**

Sheet: /Buck 5V PI/
File: Buck_5V_6A.kicad_sch

***Title: Buck converter – reusable sheet***

| Size: A4 | Date: 2026-10-06 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/15 |



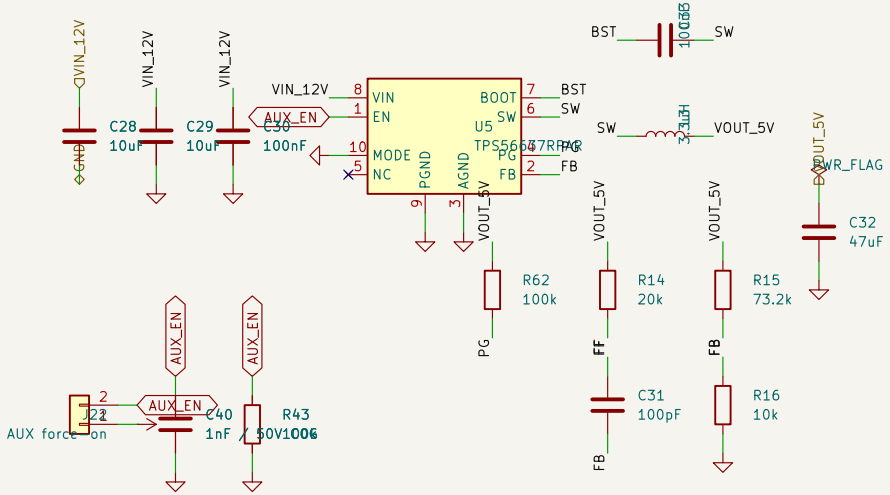

TPS54202H (U3) 12 V to 5 V 2 A, values from TI WEBENCH (Ren enables EN)

**Senior Design**

Sheet: /Buck 5V SYS/
File: Buck_5V_2A.kicad_sch

***Title: Buck converter – reusable sheet***

| Size: A4 | Date: 2026-10-06 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/15 |




TPS56637 (U5) 12 V to 5 V 6 A AUX, values from TI WEBENCH; EN = AUX_EN

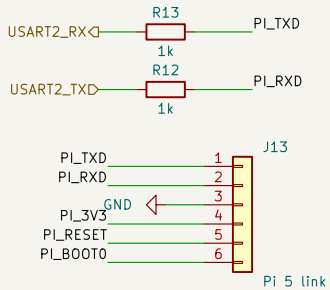
**Senior Design**

Sheet: /Buck 5V AUX/
File: Buck_5V_6A_AUX.kicad_sch

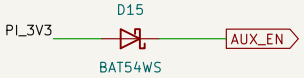
***Title: Buck converter — reusable sheet***

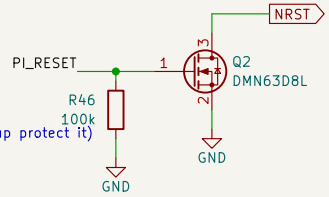
| Size: A4 | Date: 2026-10-06 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/15 |

Raspberry Pi 5 link (USART2, PA2 TX / PA3 RX), Pi 3V3 sense, STM32 reset and BOOT0

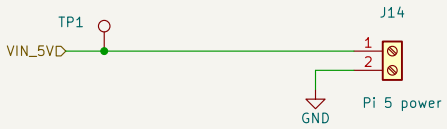



J13 1 <- Pi GPIO14 TXD
J13 2 -> Pi GPIO15 RXD
J13 3 = GND (Pi header pin 6); 3.3 V logic
R12/R13 near U1 (no ESD array fitted)




Pi 5 power: +5V_PI -> J14 USB-C pigtail (no fuse: U2's current limit and hiccup protect it)




Bring-up: leave J14 unwired, check TP1; load-test J14;
then connect the Pi. 5-6 A path




Pi-controlled functions (J13 4 <- Pi 3V3 pin 1, 5 <- GPIO23 pin 16, 6 <- GPIO24 pin 18)
AUX_EN: the +5V_AUX buck (U5) runs only while the Pi's 3V3 is up.
PI_RESET + PI_BOOT0 -> reflash the STM32 from the Pi over USART2 (AN2606:
PA2/PA3 is a bootloader port on STM32F09x). Sequence: BOOT0 high, pulse reset,
run stm32flash at 115200 8E1, then BOOT0 low and reset.
At Pi boot GPIO23/24 default to pull-down inputs, so the STM32 runs normally.

**Senior Design**

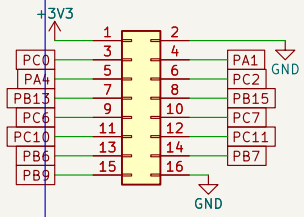
Sheet: /Pi 5 Interface/
File: Pi5_Interface.kicad_sch

***Title: Raspberry Pi 5 interface — UART and power***

| Size: A4 | Date: 2026-10-07 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 6/15 |

MCU expansion header J8 — 2.54 mm male pins, spare MCU pins only
SPI2: PB13 SCK, PC2 MISO, PB15 MOSI, PB9 CS  |  I2C1: PB6 SCL, PB7 SDA (no pull-ups on board)
USART3: PC10 TX, PC11 RX  |  ADC: PC0, PA1, PA4 (DAC)  |  PWM: PC6, PC7 (TIM3)

J8
Expansion header



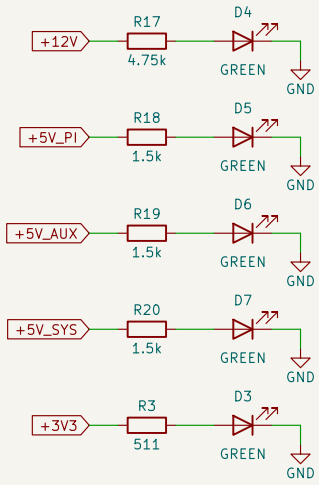

**Senior Design**

Sheet: /MCU GPIO Breakout/
File: GPIO_Breakout.kicad_sch

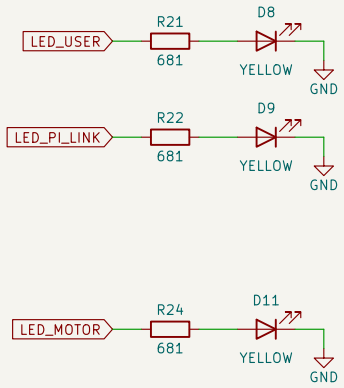
***Title: STM32 GPIO breakout headers***

| Size: A4 | Date: 2026-10-07 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 7/15 |

Power rail indicators — one green LED per rail
(Lite-On LTST-C190KGKT, 0603, Vf 2.0 V typ)




Debug LEDs — STM32 GPIO, active high (Lite-On LTST-C190KSKT yellow, 0603)




PA5 LED_USER: heartbeat
PB12 LED_PI_LINK: Pi UART activity
PB14 LED_MOTOR: motor running / fault
681R: (3.3 − 2.1) / 681 = 1.8 mA per pin

LED current ~2–2.7 mA: R = (Vrail − 2.0 V) / I. +12V uses 4.75k for the 10.8–14.6 V battery range.
All resistors 0805; worst-case dissipation 34 mW (R17 at 14.6 V).

**Senior Design**

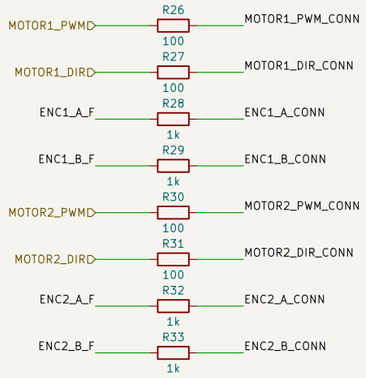
Sheet: /Power Indicators/
File: Power_Indicators.kicad_sch

***Title: Power rail and debug indicator LEDs***

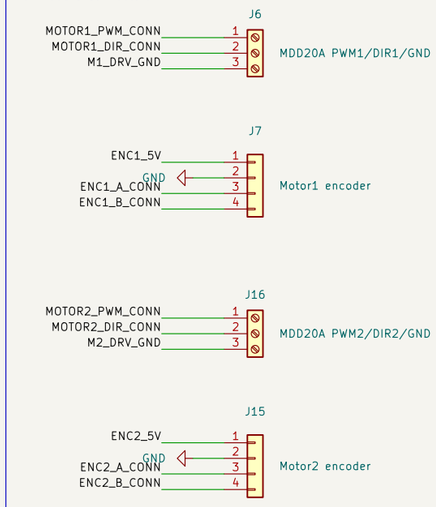
| Size: A4 | Date: 2026-10-07 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 8/15 |

Motor control — Cytron MDD20A dual-channel driver + two AndyMark PG27 motors with am-3651 encoders

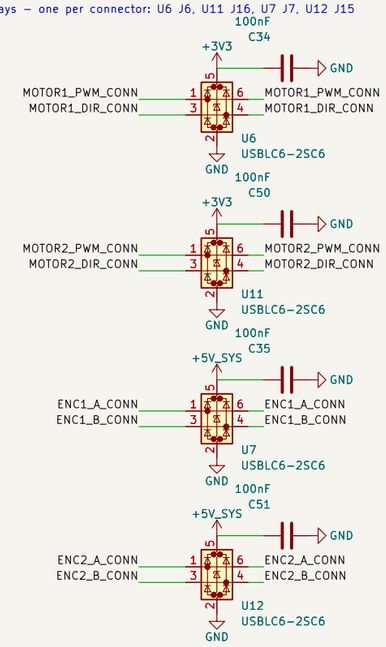
Series resistors (MCU side): 100R driver lines, 1k encoder lines



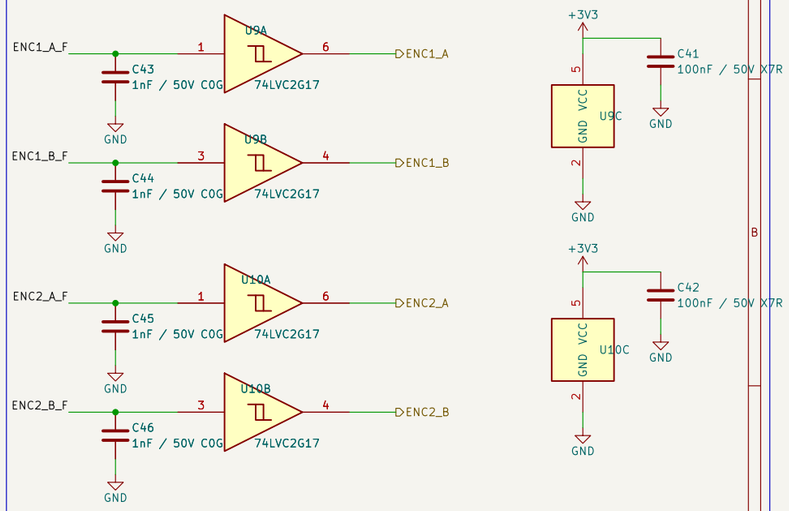

Off-board connectors



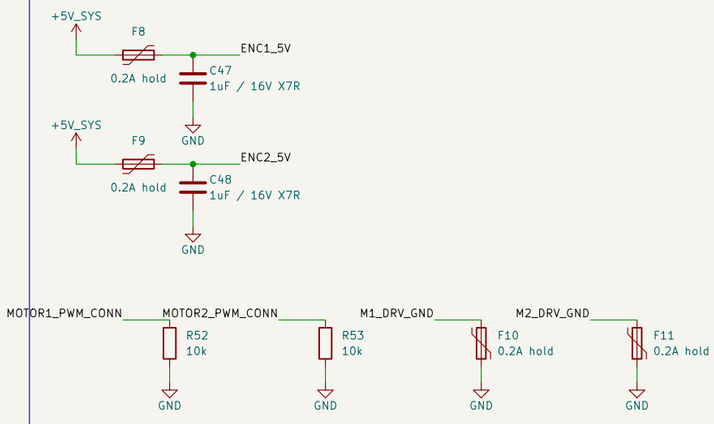

ESD arrays — one per connector: U6 J6, U11 J16, U7 J7, U12 J15




Encoder input buffers — 74LVC2G17 on +3V3 (inputs tolerate 5.5 V, even unpowered)




Encoder supply and motor-driver line protection
(If the am-3651 outputs turn out to be open-collector, bodge 4.7k from each A/B line to its ENCn_5V.)



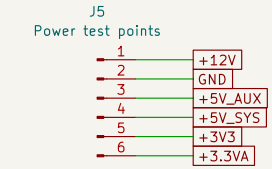

STM32F091 PIN ASSIGNMENT

| | | | | | |
|---|---|---|---|---|---|
| MOTOR1_PWM | PA6 | TIM3_CH1 | R26 100R | J6 pin 1 -> MDD20A PWM1 (R52 10k pull-down) | |
| MOTOR1_DIR | PB0 | GPIO | R27 100R | J6 pin 2 -> MDD20A DIR1 | |
| ENC1_A | PA15 | TIM2_CH1 | U9 <- R28 1k | J7 pin 3 <- encoder A | |
| ENC1_B | PB3 | TIM2_CH2 | U9 <- R29 1k | J7 pin 4 <- encoder B | |
| MOTOR2_PWM | PA7 | TIM3_CH2 | R30 100R | J16 pin 1 -> MDD20A PWM2 (R53 10k pull-down) | |
| MOTOR2_DIR | PB2 | GPIO | R31 100R | J16 pin 2 -> MDD20A DIR2 | |
| ENC2_A | PA8 | TIM1_CH1 | U10 <- R32 1k | J15 pin 3 <- encoder A | |
| ENC2_B | PA9 | TIM1_CH2 | U10 <- R33 1k | J15 pin 4 <- encoder B | |

Timers: TIM3 = both PWMs (<= 20 kHz for MDD20A); TIM2 = encoder 1, TIM1 = encoder 2.
Encoders: 5 V parts. The +3V3 rail comes up after +5V_SYS, and ST allows only 3.6 V on any pin while VDD = 0 (AN4899), so A/B go through U9/U10 (74LVC2G17 on +3V3) instead of straight to the MCU. Never enable internal pull-ups on these pins.
PWM pull-downs: MDD20A brakes on PWM low, so a resetting MCU always means 'brake'.
J6/J16 pin 3 is a signal reference only (through F10/F11). Battery and motor leads wire directly to the MDD20A.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Motor Control/ | | |
| File: Motor_Control.kicad_sch | | |
| ***Title: Motor control — MDD20A driver and encoders*** | | |
| Size: A3 | Date: 2026-10-08 | **Rev: 1.1** |
| KiCad E.D.A. 10.0.6 | | Id: 9/15 |

Power test points




J5: one 2.54 mm header for every rail
1 +12V (after Q1)
2 GND
3 +5V_AUX
4 +5V_SYS
5 +3V3
6 +3.3VA (MCU analog)
+12V is next to GND only.




TP3: GND loop for the scope ground clip.
+5V_PI is on TP1 (Pi 5 Interface sheet),
used in the Pi bring-up check.

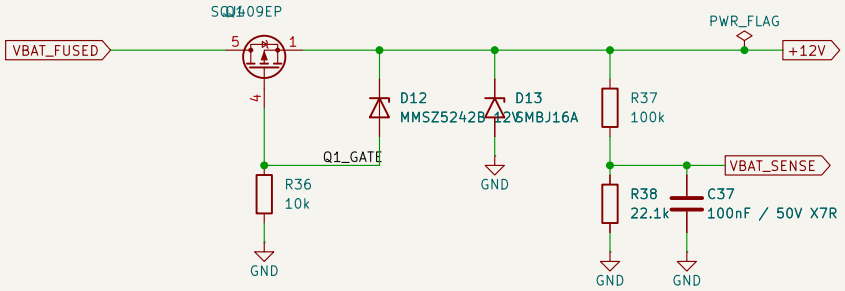
**Senior Design**

Sheet: /Test Points/
File: Test_Points.kicad_sch

***Title: Power test points***

| Size: A4 | Date: 2026-10-08 | **Rev: 1.1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 10/15 |

Input protection — reverse-polarity P-FET, gate clamp, TVS, battery sense



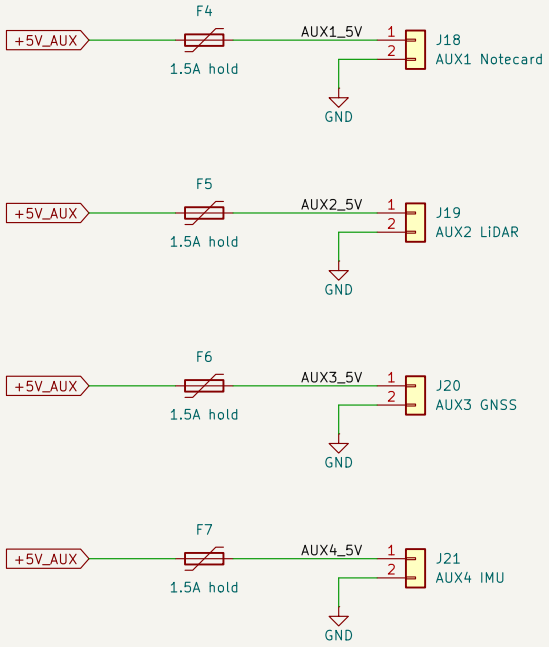

Battery chain (root sheet): J2 XT30 -> VBAT_IN -> F1 10 A MINI fuse -> VBAT_FUSED -> this sheet -> +12V.
C36 (220 uF bulk, root sheet) is on +12V, after Q1, so a reversed battery can never reverse-bias it.

Q1: high-side P-FET. A reversed battery leaves the body diode reverse-biased and VGS = 0, so nothing downstream of Q1 ever sees negative voltage. Forward: about 7 mOhm, 0.2 W at 5 A.
D12 limits |VGS| to 12 V (part limit 20 V); R36 holds the gate at GND.
D13 (SMBJ16A) clamps fast spikes on +12V to about 26 V. It is NOT sized for motor regen if the BMS disconnects (about 20 J); that must be clamped on the motor-driver side. If D13 fails short, F1 opens.

VBAT_SENSE -> PA0 (ADC_IN0): VBAT = V(PA0) x 122.1 / 22.1 (5.525 x). 10.0 V -> 1.81 V, 14.6 V -> 2.64 V.
Layout: Q1, D12, D13 between F1 and C36; D13 next to C36 with short, wide GND.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Input Protection/ | | |
| File: Input_Protection.kicad_sch | | |
| ***Title: Input protection and battery sense*** | | |
| Size: A4 | Date: 2026-10-15 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 11/15 |

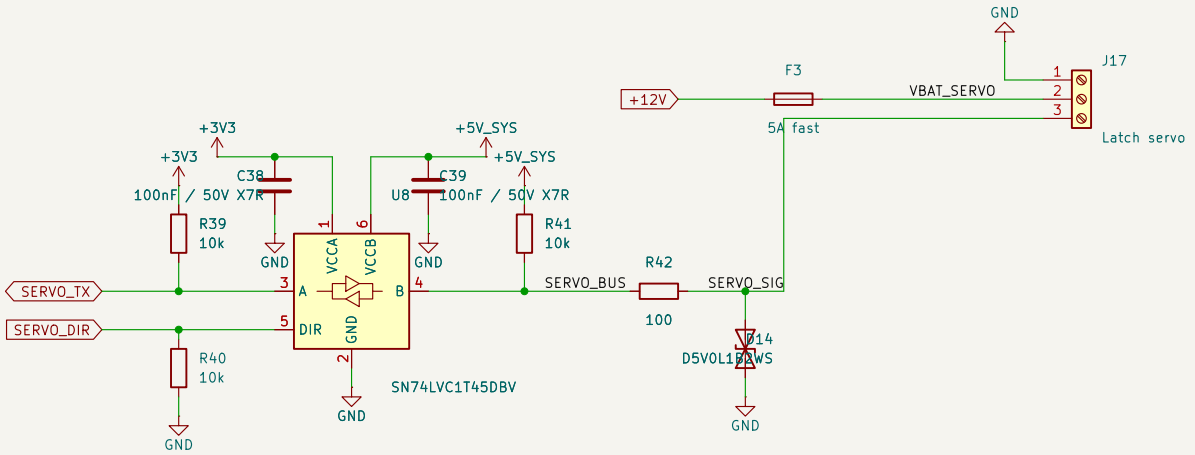
## +5V_AUX peripheral outputs — four ports, one PTC each




+5V_AUX comes from U5 (TPSM63606, 6 A), which is enabled when the Pi 5 is up (AUX_EN, see Buck 5V AUX).
Each port has its own PTC so a pinched or shorted peripheral cable only drops that port.
Connectors: JST XH 2.5 mm top-entry (plug in from above), keyed, 3 A per contact.
Use pre-made 2-pin XH cables; solder or splice the free end at each peripheral. Pin 1 = +5 V, pin 2 = GND.
PTC hold is 1.5 A on every port (trip about 3 A) so one part covers all four; bursts of 2 A for
milliseconds (Notecard modem) do not trip it. Check each peripheral's peak current before power-up.
Layout: ports on one board edge next to U5; PTCs at the connectors; +5V_AUX on a pour.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /AUX Outputs/ | | |
| File: AUX_Outputs.kicad_sch | | |
| ***Title: +5V_AUX peripheral outputs*** | | |
| Size: A4 | Date: 2026-10-15 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 12/15 |

# Latch servo interface — Hiwonder HTS-30HS serial bus servo



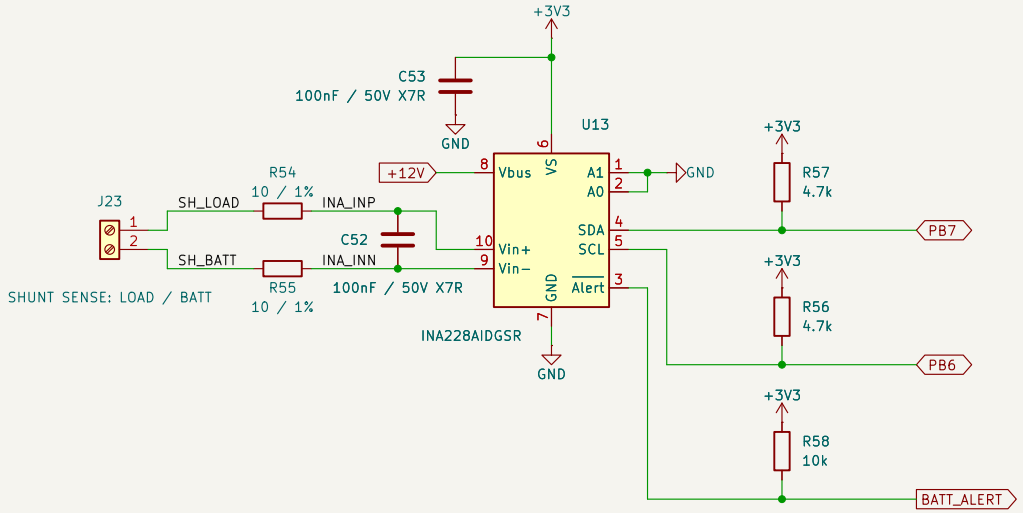

HTS-30HS: 9–12.6 V (rated 11.1 V), 100 mA no-load, 3.2 A stall, half-duplex UART 115200 baud, packets start 0x55 0x55.
Power: a full 4S LiFePO4 is 14.6 V, above the servo's 12.6 V maximum, so J17 pin 2 feeds an EXTERNAL buck module set to 11.1–12 V (rated 5 A or more). Its output and GND go to the servo plug; SERVO_SIG goes straight to the plug.
Do not also tie the buck or servo ground to the battery or chassis: the servo current must return only through J17 pin 1.

Signal: PB10 = USART3_TX (AF4) in single-wire half-duplex mode (HDSEL), configured open-drain with R39 pull-up.
PB11 = SERVO_DIR: drive high just before sending, low as soon as the last byte's TC flag sets (the servo answers read commands immediately). U8 translates 3.3 V <-> 5 V; Hiwonder's reference bus has a 10k pull-up to 5 V (R41).
Firmware note: in half-duplex mode the USART also receives its own transmitted bytes; discard them.

| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Servo Interface/ | | |
| File: Servo_Interface.kicad_sch | | |
| ***Title: Latch servo interface (Hiwonder HTS-30HS)*** | | |
| Size: A4 | Date: 2026-10-15 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 13/15 |

# Battery current monitor — INA228 on an off-board 50 A / 75 mV shunt

U13 (INA228) measures battery current on an OFF-BOARD 50 A / 75 mV shunt (1.5 mOhm, class 0.5) in the battery NEGATIVE lead.
Motor current never flows through this board (2.13). Shunt lugs: battery − on one side; the robot's common ground lug
(J2 pin 1 harness lead, MDD20A VB−, charger −) on the other. Everything that returns to battery − must pass through the shunt.
J23: twisted 22–24 AWG Kelvin pair from the shunt's small sense screws, routed away from the motor leads.
Pin 1 = LOAD (ground-lug side) -> IN+, pin 2 = BATT (battery − side) -> IN−. Positive current = discharge.
Low side: the sense wires sit within about 0.1 V of GND, so a chafed sense wire cannot short the battery.
Range: board GND sits up to ~60 mV above the shunt, so IN− is about −135 mV at 50 A (limit −0.3 V).
Full scale (ADCRANGE = 0) is +/−163.84 mV = +/−109 A. R54/R55 10 ohm + C52: input filter and dV/dt protection (SLYS021A 8.1.4).
No ground connection at the inputs: IN+ must NOT tie to board GND on the board.
VBUS reads +12V (after Q1, so never reversed; D13 limits it to 26 V, 85 V rating). I2C1 on PB6/PB7, address 0x40 (A0 = A1 = GND).
ALERT is open-drain, active low -> PB4 (EXTI4). Firmware: SHUNT_CAL = 13107.2e6 x 250e−6 x 0.0015 = 4915 (CURRENT_LSB = 250 uA).
Layout: U13, R54, R55, C52 next to J23 on an edge away from U2/U5 and L1; INA_INP/INA_INN as a tight pair, no vias, no GND pour.




**Senior Design**

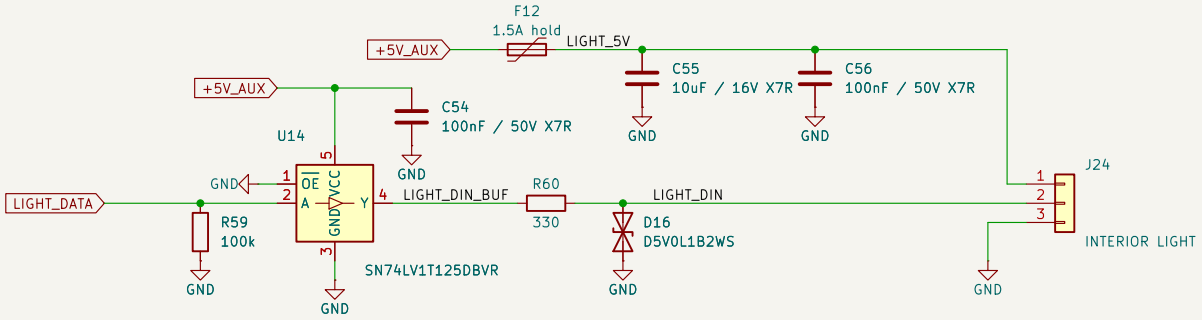
Sheet: /Battery Monitor/
File: Battery_Monitor.kicad_sch

***Title: Battery current monitor (INA228, off-board shunt)***

| Size: A4 | Date: 2026-10-27 | **Rev: 1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 14/15 |

# Payload compartment interior light — WS2812B strip (S17.2)

S17.2: the light turns on within 0.5 s of the lid opening and off within 2 s of it closing. Off-board WS2812B strip, up to 16 LEDs;
firmware caps its draw at 0.5 A. Power: +5V_AUX through F12 (1.5 A PTC, same part as the AUX ports), so the light is only powered
while the Pi is up. Off-board: put a 470–1000 uF / 10 V electrolytic across the strip power at the strip end.
Data: PB5 = SPI1_MOSI (AF0), TX-only master with DMA, 6 MHz (48 MHz / 8); one SPI byte per WS2812 bit (0 -> 0xC0, 1 -> 0xF0);
end each frame with at least 300 us of zeros. SPI1 SCK is not mapped to a pin. Fallback: TIM16_CH1 on PB8 (AF2) + DMA (schematic change).
U14 (SN74LV1T125, VIH 2.03 V at VCC = 5 V) lifts the 3.3 V data to 5 V: WS2812B needs VIH >= 0.7 x VDD = 3.5 V.
Its input has no clamp diode to VCC (SCLS745D Fig. 8-2, VI max 7 V, II +/-1 uA at VCC = 0 V), so PB5 may drive it while +5V_AUX is off.
R59 holds the data low while the MCU resets. R60 damps the cable; D16 is ESD protection at J24.
Interlock: firmware turns the light off whenever the latch servo moves (keeps the worst-case input at 8.4 A; see fuse-selection.md).
J24 (JST XH, top entry): 1 = +5 V (LIGHT_5V), 2 = DIN, 3 = GND. Layout: U14, R60, D16, C55, C56 and F12 at J24, on the payload-side edge.




| | | |
|---|---|---|
| **Senior Design** | | |
| Sheet: /Interior Light/ | | |
| File: Interior_Light.kicad_sch | | |
| ***Title: Payload compartment interior light (WS2812B)*** | | |
| Size: A4 | Date: 2026-10-27 | **Rev: 1.0** |
| KiCad E.D.A. 10.0.6 | | Id: 15/15 |